

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=other

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	other	silpo_novus	promo_controlled	nan	nan
prozorro	0.9029715543675231	0.01	1.3734510184087591e-14	0.1	1.260173359802745e-10	0.1	I(1)	1.0	other	silpo_novus	promo_controlled	nan	nan
retail	0.13774846220222936	0.1	0.022304689067361494	0.1	1.0563874553359415e-08	0.1	I(1)	0.0	other	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo_novus	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo_novus	promo_controlled	1.1627575743616856e-06	52.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	other	silpo_novus	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
other	silpo_novus	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	52	3.191807377212619	-12.700814525525912	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.001596; ... Primary chain is strictly...	
other	silpo_novus	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	50	0.7522139284224689	6.209165049284258	-0.4291476640555171	0.00407796931835764	0.859423951247572	0.4819111926141287	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	prozorro_to_retail	retail	prozorro	0.38926394797562297	0.10283762781865258	0.894300321098457	0
NARDL	prozorro_to_retail	retail	prozorro	0.23976635344270975	0.005946756007259149	6.260837456608593e-56	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-04-14	nan	5.0	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-15	nan	5.295	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-16	nan	4.985	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-17	nan	5.43	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-18	nan	5.100000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-19	nan	5.175000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-21	nan	5.4	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-22	nan	5.25	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-23	nan	5.0600000000000005	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-24	nan	5.235	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-25	nan	5.46	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-28	nan	5.2	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-29	nan	4.72	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-04-30	nan	4.775	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-01	nan	5.08	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-02	nan	5.05	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-05	nan	5.100000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-06	nan	4.995	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-07	nan	5.22	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-08	nan	5.04	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-09	nan	4.675000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-12	nan	4.78	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-13	nan	4.890000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-14	nan	5.2	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-15	nan	4.95	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-16	nan	5.04	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-19	nan	4.715	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-20	nan	4.545	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-21	nan	4.59	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-22	nan	4.77	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-23	nan	5.265000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-25	nan	4.5	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-26	nan	5.33	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-27	nan	5.55	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-28	nan	4.56	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-29	nan	4.98	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-05-30	nan	4.65	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-02	nan	4.66	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-03	nan	4.700000000000001	nan	other	silpo_novus	promo_controlled	daily_median_of_available...
2025-06-04	nan	4.7	nan	other	silpo_novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
prozorro_to_retail	1	0.021801623475124304	
prozorro_to_retail	2	-0.1677317854495336	
prozorro_to_retail	3	-0.18614997912727976	
prozorro_to_retail	4	-0.11954773824154614	
prozorro_to_retail	5	-0.3024125191964572	
prozorro_to_retail	6	-0.3584160123905419	
prozorro_to_retail	7	-0.256406254168928	
prozorro_to_retail	8	-0.15845877042218173	
prozorro_to_retail	9	-0.3378266814360856	
prozorro_to_retail	10	-0.15596459462221665	
prozorro_to_retail	11	-0.15873386433990713	
prozorro_to_retail	12	-0.244078537874973	
prozorro_to_retail	13	-0.2191276584924703	
prozorro_to_retail	14	-0.09053412649840303	
prozorro_to_retail	15	-0.019377145758522434	
prozorro_to_retail	16	-0.028522233066124413	
prozorro_to_retail	17	-0.011156583097595032	
prozorro_to_retail	18	-0.05490922876379749	
prozorro_to_retail	19	-0.20245354984748348	
prozorro_to_retail	20	-0.13694972856526602	
prozorro_to_retail	21	-0.09355059469842485	
prozorro_to_retail	22	0.024867173761834425	
prozorro_to_retail	23	0.039988247700322485	
prozorro_to_retail	24	0.02962642392675777	
prozorro_to_retail	25	-0.022052301936377822	
prozorro_to_retail	26	-0.08309220249709504	
prozorro_to_retail	27	-0.16472583140664881	
prozorro_to_retail	28	-0.16597401392858518	
prozorro_to_retail	29	-0.12394813069095229	
prozorro_to_retail	30	-0.17429367595355302	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
other	silpo_novus	promo_controlled	daily	ambiguous	I(1)	I(1)	0	232	56	0	52	0	Low pair overlap in main ...

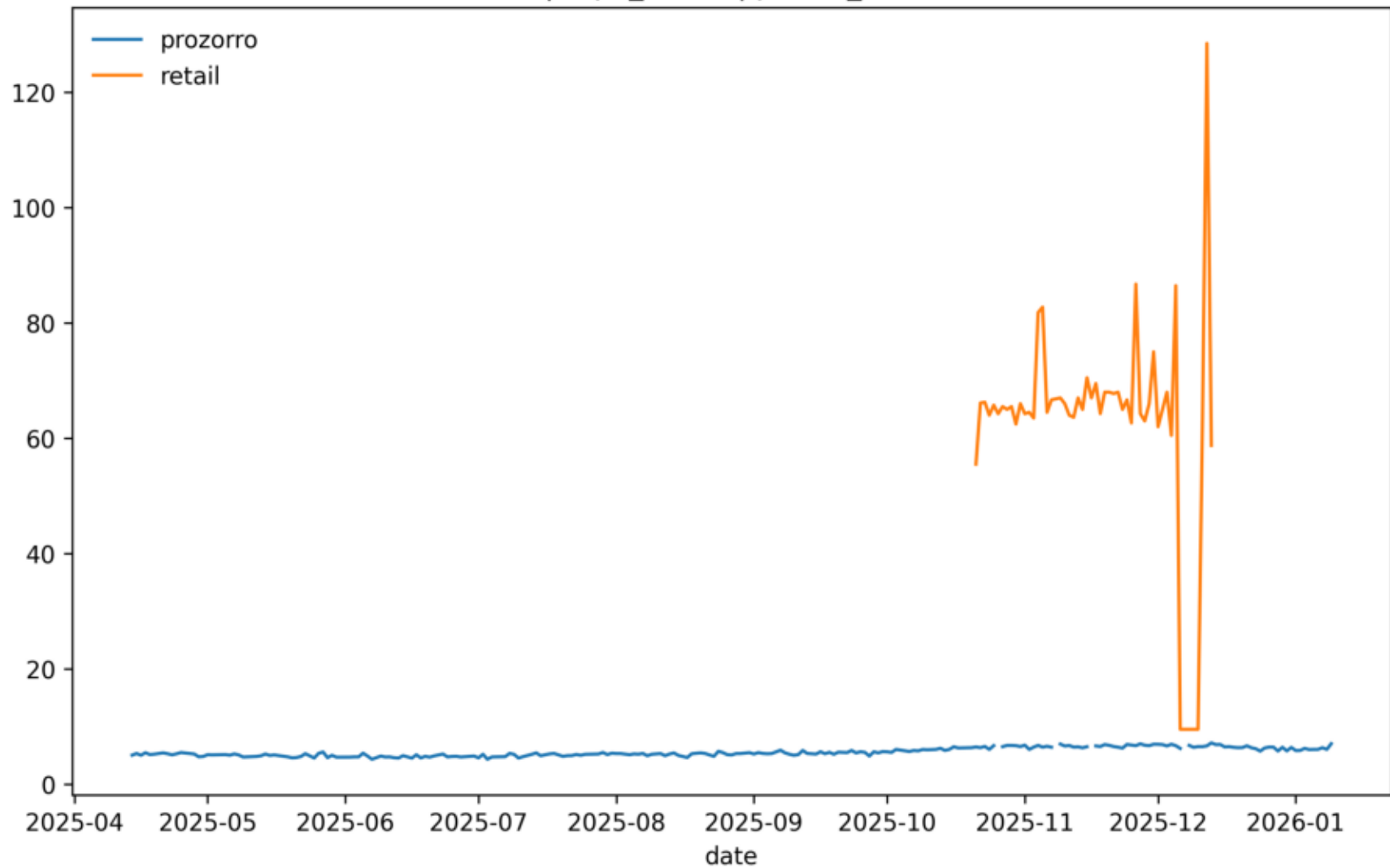
Table: NARDL_Multipliers

horizon	mult_pos		mult_neg	link
0	-1.2056182554944759	-1.9578321839169448	prozorro_to_retail	
1	-1.3625211573833618	-2.21263054124775	prozorro_to_retail	
2	-1.3829409880461419	-2.2457907903394725	prozorro_to_retail	
3	-1.3855984881897139	-2.250106368082376	prozorro_to_retail	
4	-1.3859443435070138	-2.25066801076496	prozorro_to_retail	
5	-1.3859893541915076	-2.2507411046869543	prozorro_to_retail	
6	-1.3859952120216963	-2.250750617356839	prozorro_to_retail	
7	-1.3859959743778993	-2.250751855365224	prozorro_to_retail	
8	-1.3859960735933017	-2.250752016483478	prozorro_to_retail	
9	-1.3859960865055023	-2.2507520374519077	prozorro_to_retail	
10	-1.3859960881859361	-2.2507520401808043	prozorro_to_retail	
11	-1.385996088404633	-2.2507520405359513	prozorro_to_retail	
12	-1.3859960884330949	-2.2507520405821713	prozorro_to_retail	
13	-1.385996088436799	-2.2507520405881865	prozorro_to_retail	
14	-1.385996088437281	-2.2507520405889694	prozorro_to_retail	
15	-1.3859960884373437	-2.250752040589071	prozorro_to_retail	
16	-1.385996088437352	-2.2507520405890844	prozorro_to_retail	
17	-1.385996088437353	-2.250752040589086	prozorro_to_retail	
18	-1.3859960884373532	-2.250752040589086	prozorro_to_retail	
19	-1.3859960884373532	-2.250752040589086	prozorro_to_retail	
20	-1.3859960884373532	-2.250752040589086	prozorro_to_retail	

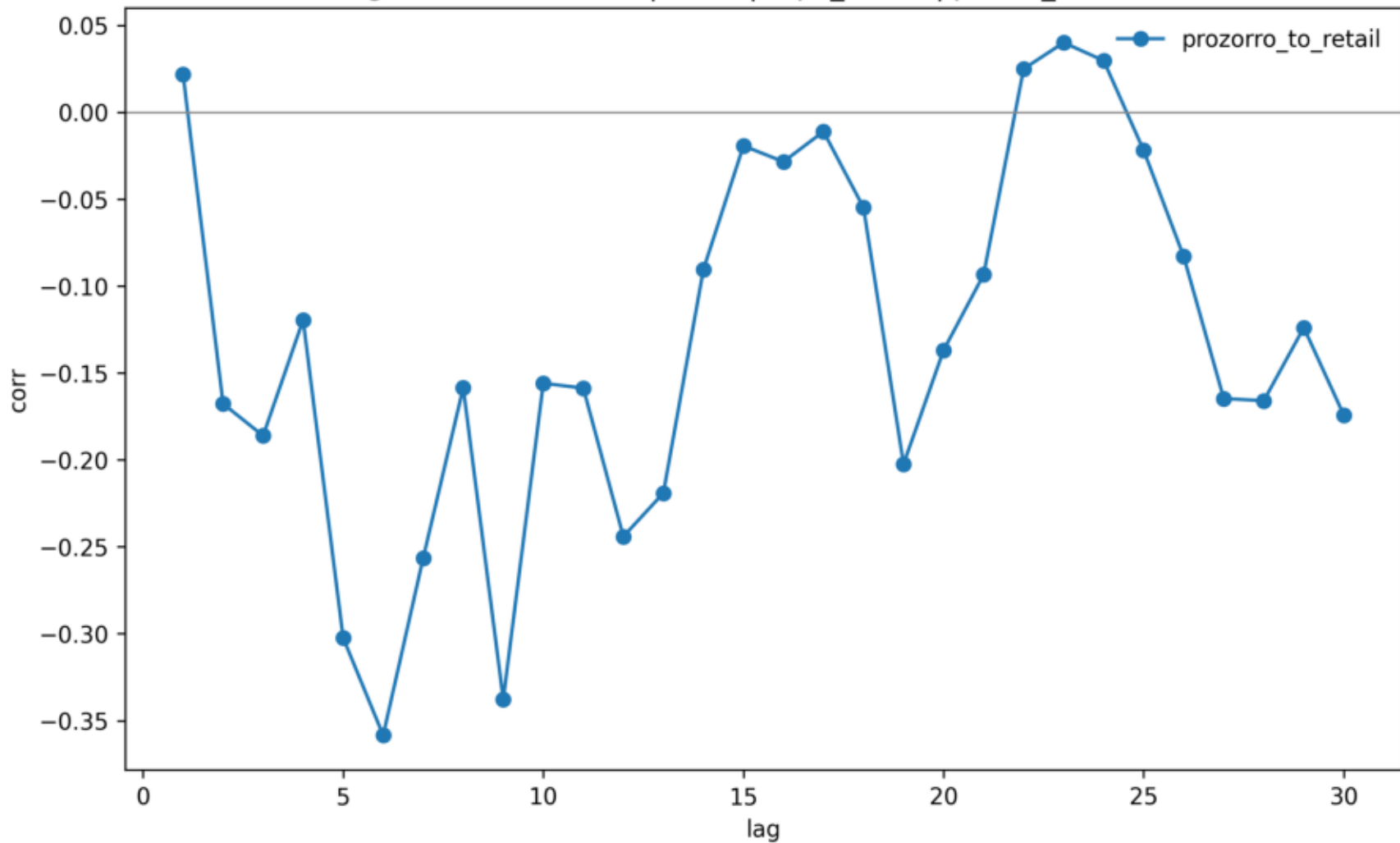
Table: VECM_IRF

No rows

other | silpo_novus | promo_controlled



Lag-Correlation Profile | other | silpo_novus | promo_controlled



NARDL Dynamic Multipliers | other | silpo_novus | promo_controlled

